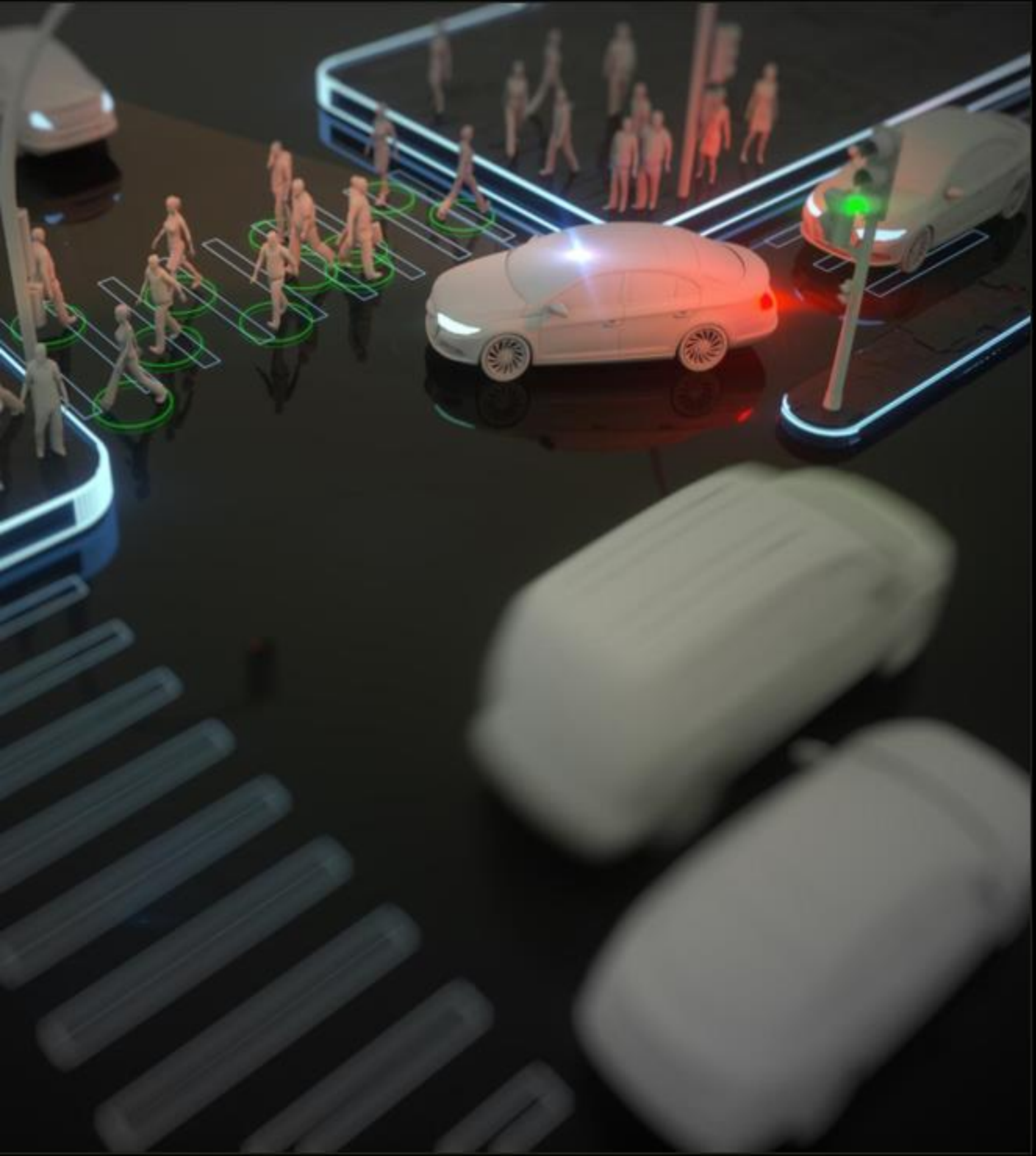


PREDICTING VEHICLE COLLISION CAUSES

Analyzing trends and factors behind recent vehicle accidents

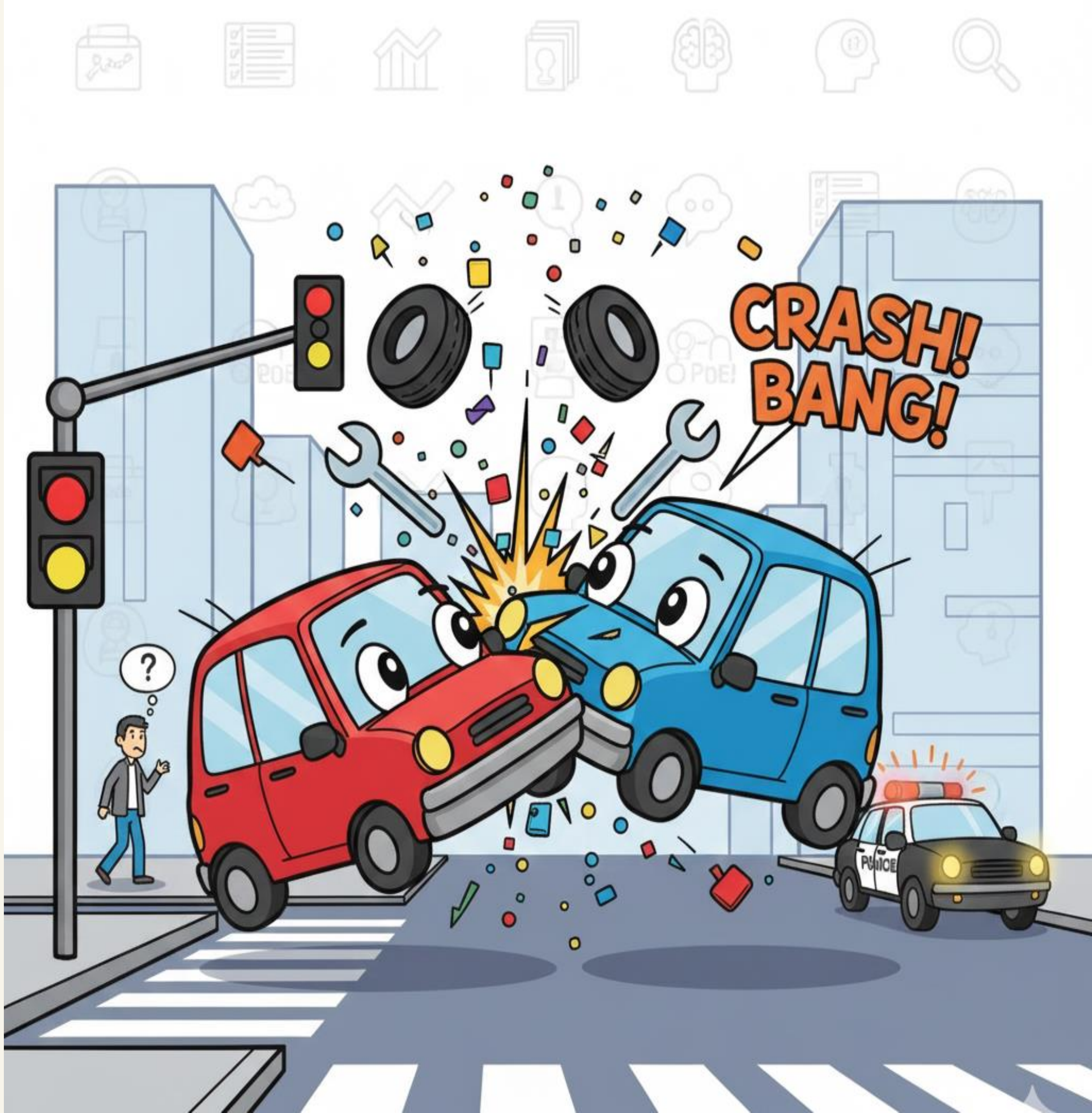
Vandan, Shiva, Utkarsh

INTRODUCTION

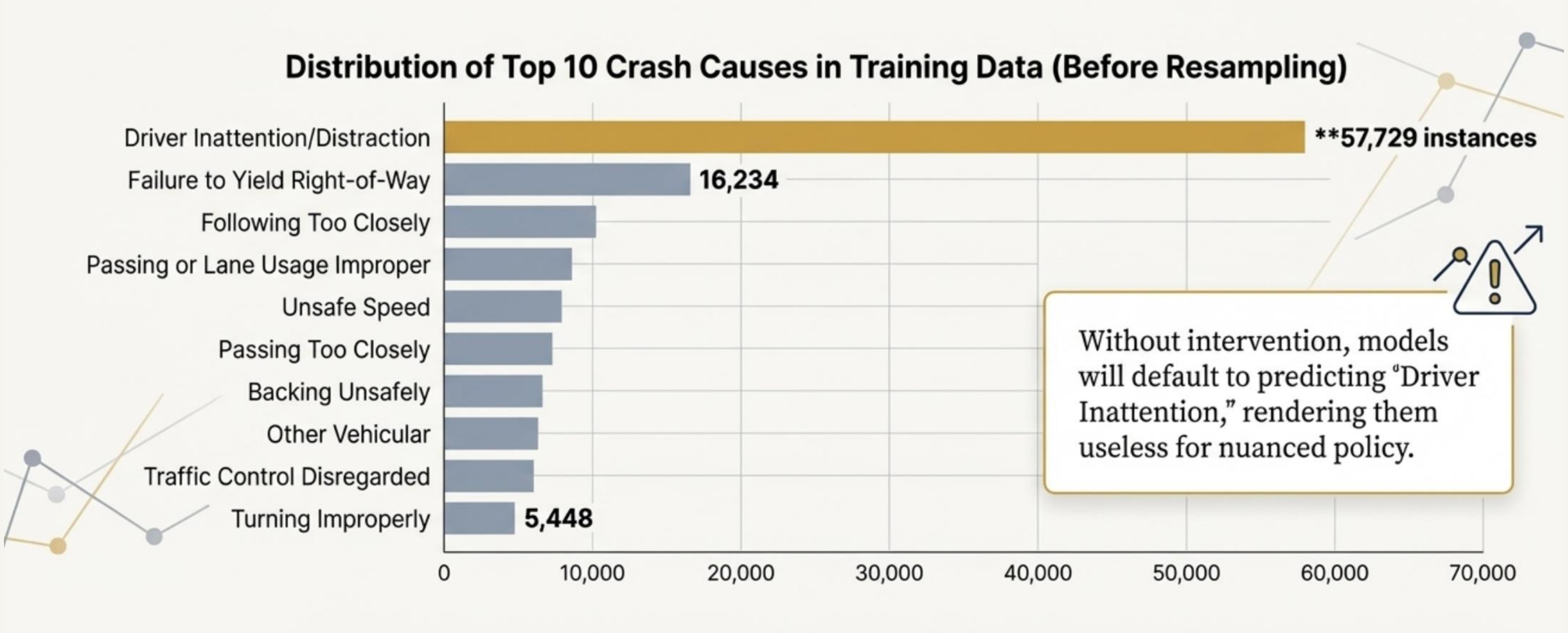
PROBLEM STATEMENT

Moving beyond simple counts to causal classification is a significant data challenge.

- **High-Volume & Irregular:** Collision datasets are massive (over 2.2 million records) but often contain sparse, inconsistent, and missing metadata.
- **Limited Actionable Insights:** Raw counts like "Brooklyn has the most accidents" are descriptive but don't explain why, hindering targeted safety interventions.
- **The True Goal:** The objective isn't just to count incidents, but to accurately classify the primary **contributing factor** for each collision.



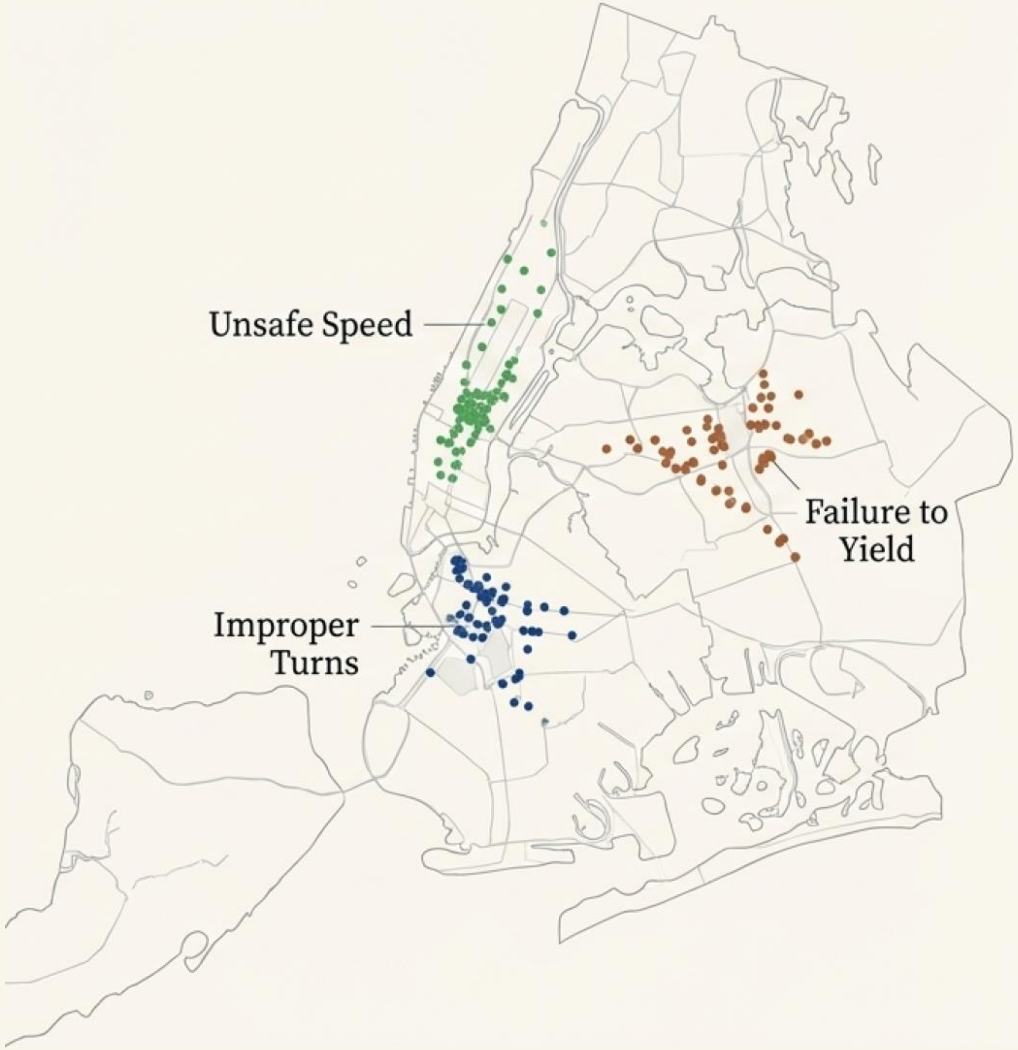
CORE CHALLENGE: SEVERE CLASS IMBALANCE



WHY THIS MATTERS : THE IMPACT ON PUBLIC SAFETY

Accurate, granular etiology unlocks smarter, more effective safety strategies.

- **Targeted Interventions:** Differentiate between problems requiring engineering fixes (e.g., poor intersection design) and those needing behavioral enforcement (e.g., speeding).
- **Local Optimization:** Generate borough- or street-specific insights. Is "**Failure to Yield**" a bigger problem in Queens than in Manhattan?
- **Informed Policy:** Provide data-driven evidence to support urban planning decisions, traffic law adjustments, and public safety campaigns.



OBJECTIVES: OUR THREE-PRONGED APPROACH

1. Class Rebalancing

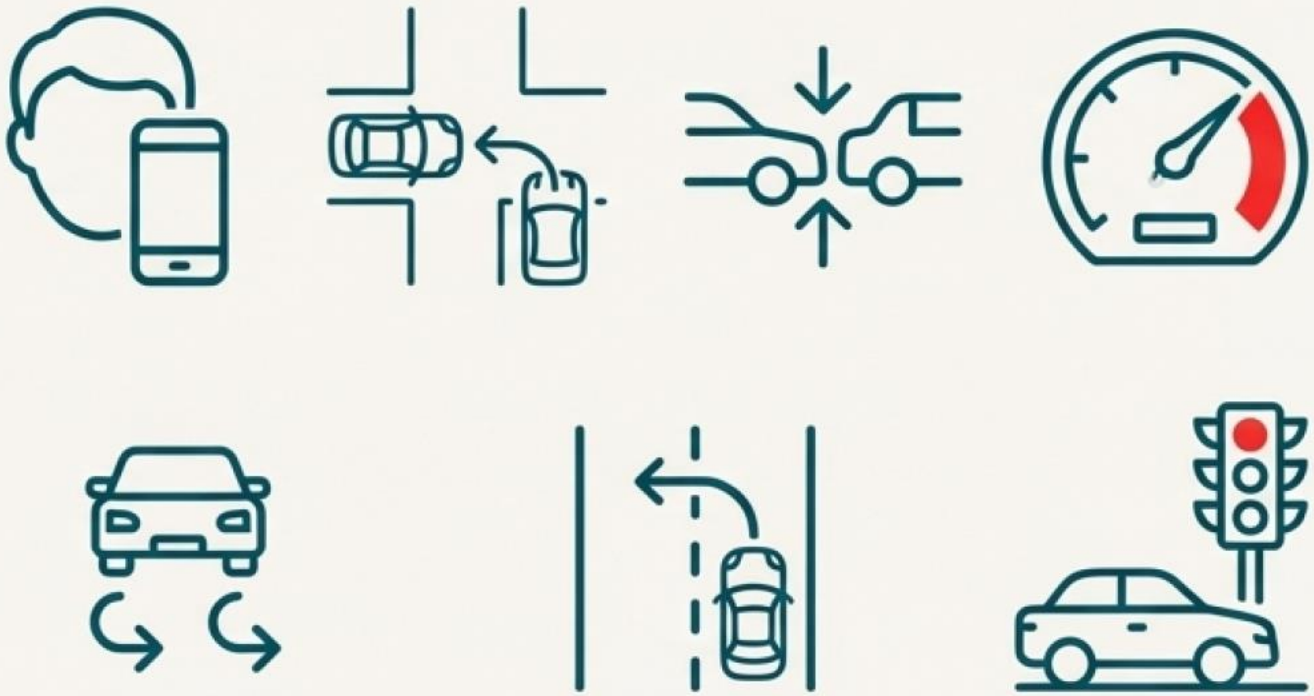
- Undersample majority
- Oversample minority
- Improve sensitivity to rare causes

2. Predictive Modeling

- Multi-class classification
- Algorithm: **LightGBM**
- Inputs: time, location, vehicle type

3. Explainability (XAI)

- SHAP (global + local)
- LIME (per-instance attribution)
- Transparency for stakeholders



DATASET

Sourcing and Scoping the Crash Data

Data Source

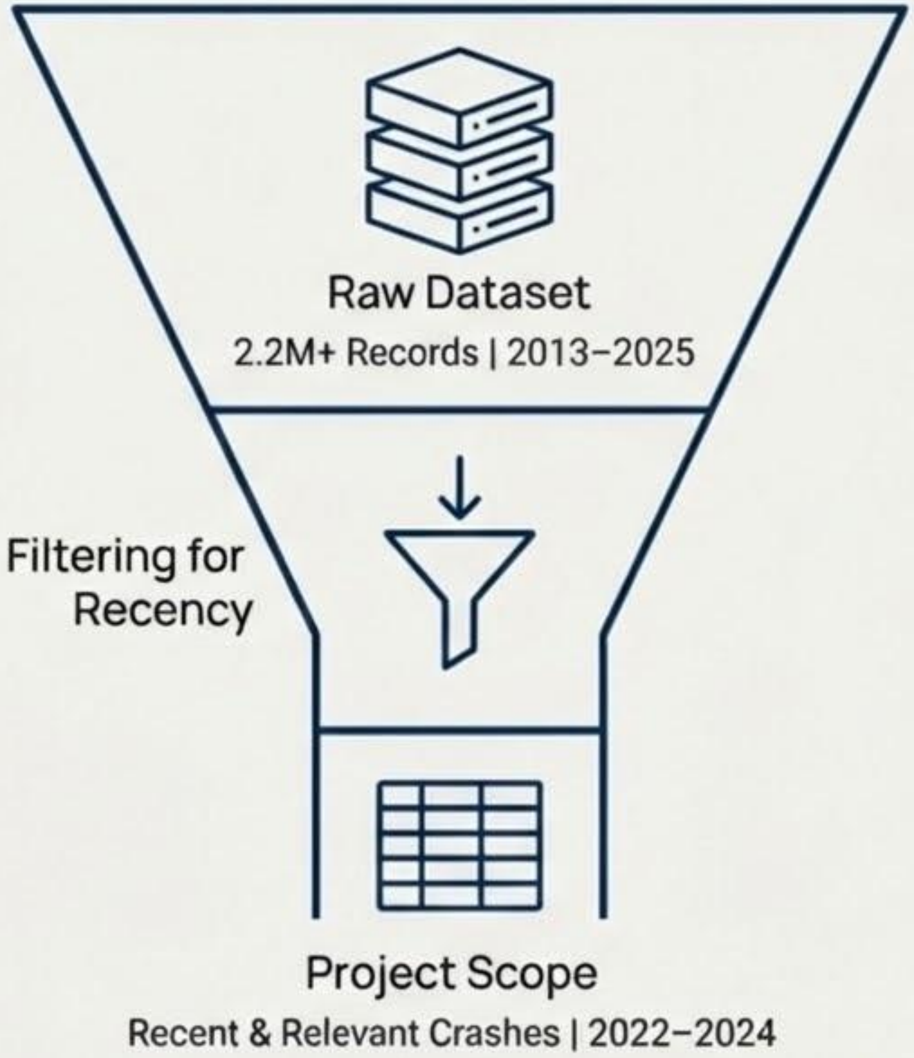
U.S. Government open data portal, featuring the comprehensive “Motor Vehicle Collisions – Crashes” dataset.

Original Scale

The raw dataset is massive, containing **2,216,469** and **29** crash records and distinct columns. The original time frame spanned from 2013 to mid-2025.

Project Scope

To ensure recency and relevance for predictive modeling, our analysis is filtered to crashes occurring in **2022, 2023, and 2024.**



KEY PATTERNS

Hourly Trends

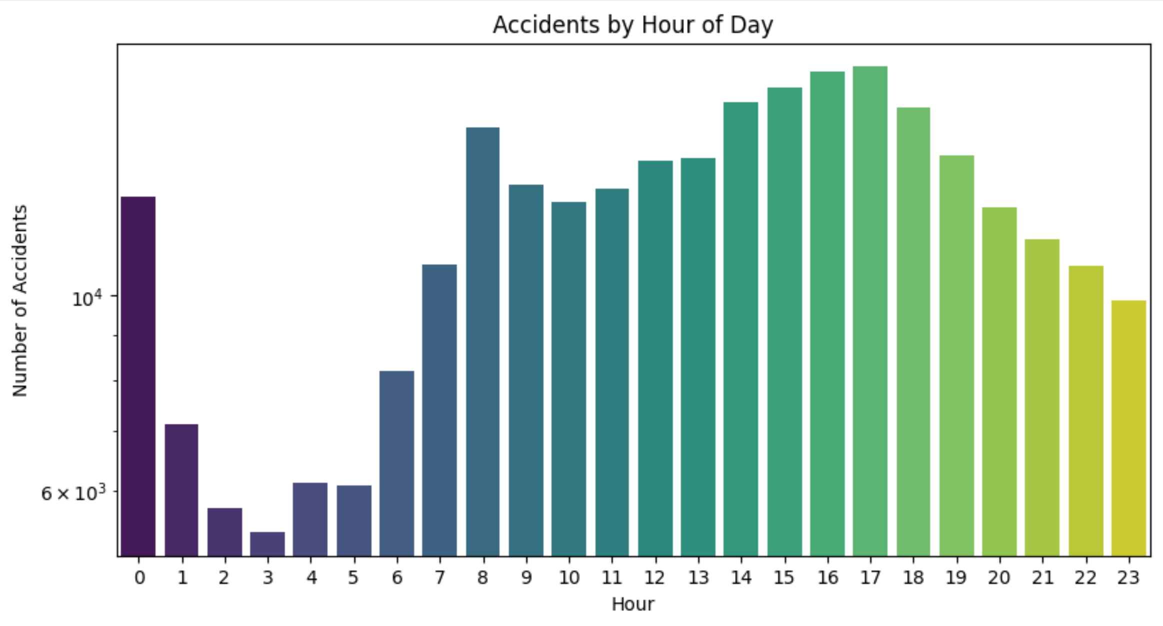
Accidents show a clear daily rhythm, dipping in the early morning (2-5 AM) and peaking during morning (8-9 AM) and evening (3-6 PM) commutes.

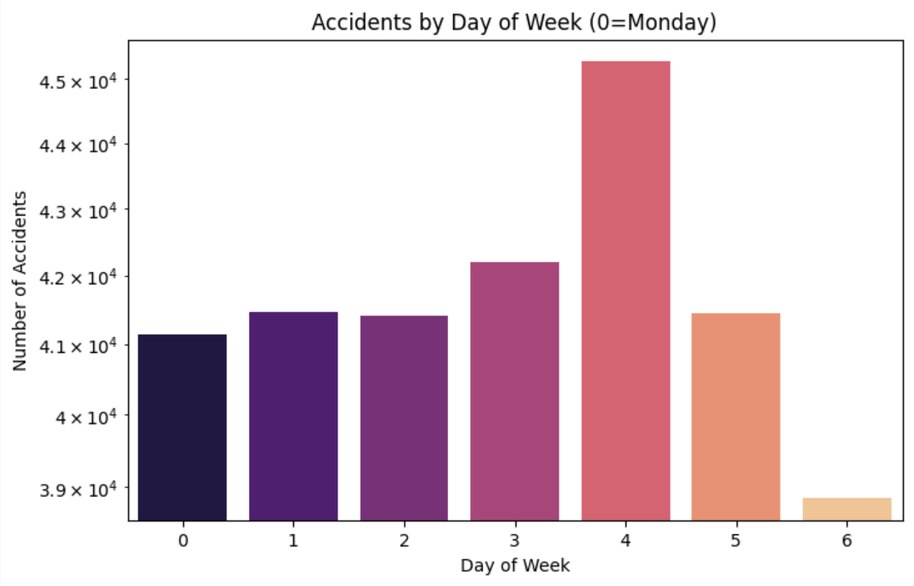
Weekly Trends

Collisions are consistently high on weekdays and drop significantly over the weekend, especially on Sunday, reflecting commuter traffic patterns.

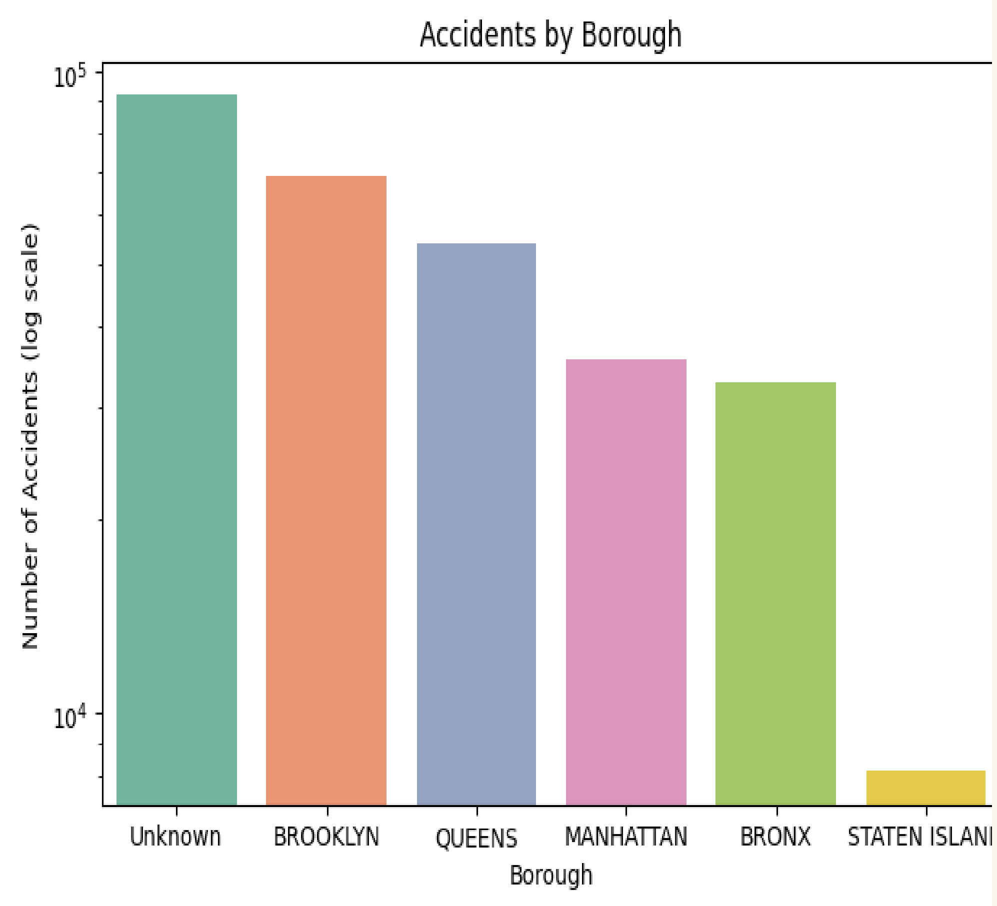
Geographic Distribution

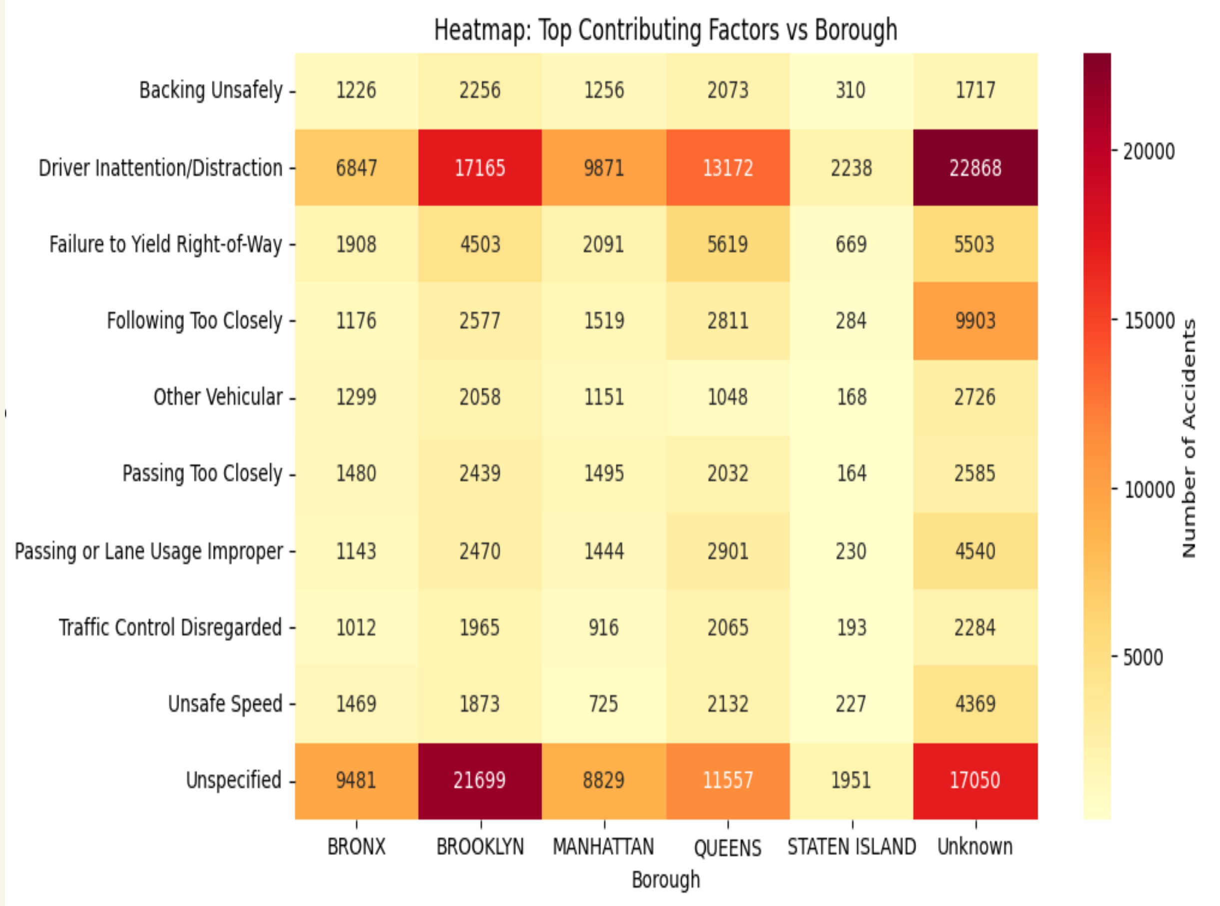
Among specified boroughs, Brooklyn and Queens record the highest number of accidents. A large number of records have an 'Unknown' borough, which is retained for its non-spatial features.





KEY PATTERNS





	BRONX	BROOKLYN	MANHATTAN	QUEENS	STATEN ISLAND	Unknown
Backing Unsafely	1226	2256	1256	2073	310	1717
Driver Inattention/Distraction	6847	17165	9871	13172	2238	22868
Failure to Yield Right-of-Way	1908	4503	2091	5619	669	5503
Following Too Closely	1176	2577	1519	2811	284	9903
Other Vehicular	1299	2058	1151	1048	168	2726
Passing Too Closely	1480	2439	1495	2032	164	2585
Passing or Lane Usage Improper	1143	2470	1444	2901	230	4540
Traffic Control Disregarded	1012	1965	916	2065	193	2284
Unsafe Speed	1469	1873	725	2132	227	4369
Unspecified	9481	21699	8829	11557	1951	17050

A Multi-Faceted Strategy for Balanced Learning

To address the severe class imbalance and build a robust model, we implemented a three-part strategy:

1. Focus on Actionable Classes

We filtered the dataset to the **top 10 most frequent contributing factors.** This improves model focus by removing extremely rare or ambiguous labels like 'Unspecified,' ensuring each class has sufficient data for learning.

2. Apply Strategic Resampling

We adopted a targeted resampling workflow on the training data. This involves **undersampling** the majority class and **oversampling** the minority classes to create a balanced dataset for the model to learn from.

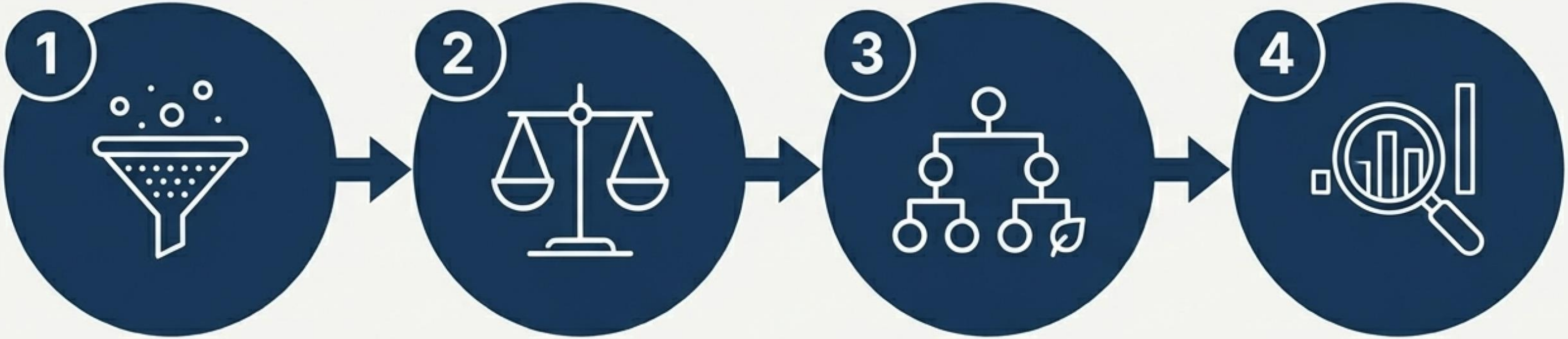
3. Evaluate with a Fair Metric

We chose the **Weighted F1-score** as the primary evaluation metric. Unlike raw accuracy, this score accounts for class distribution, providing a more reliable assessment of the model's performance across both common and rare classes.

METHODS AND SETUP

Our Methodological Blueprint for a Robust & Interpretable Model

To address the challenges of scale and imbalance, we designed a comprehensive pipeline focused on data integrity, model performance, and ultimate transparency.



Data Preparation & Filtering

Focusing the dataset on the most relevant data and impactful crash causes.

Strategic Resampling

Actively engineering a balanced training set to mitigate model bias.

LightGBM Modeling & Tuning

Training a high-performance gradient boosting model optimized for speed and accuracy.

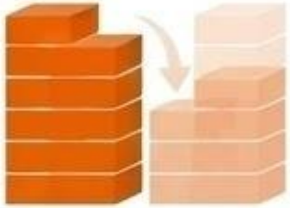
XAI-Driven Interpretation

Using SHAP and LIME to dissect model predictions and extract human-readable insights.

Step 1: Correcting Imbalance with a Hybrid Resampling Strategy

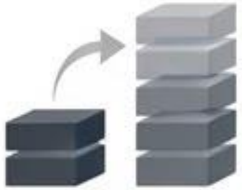
To ensure the model learns to identify minority classes effectively, we implemented a strategic hybrid resampling approach using `RandomUnderSampler` and `RandomOverSampler`.

1. Undersampling the Majority

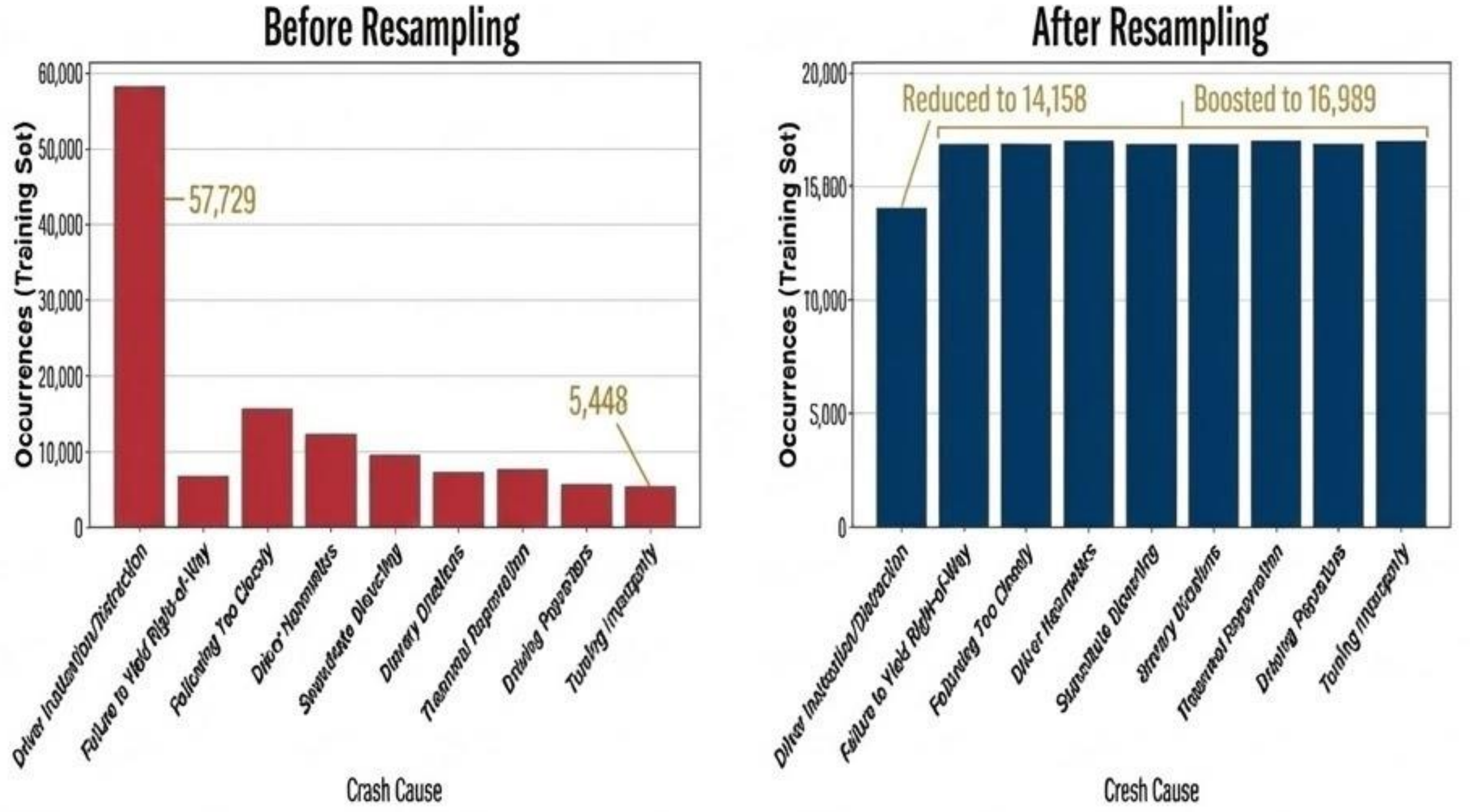


The "Driver Inattention/Distraction" class is reduced to the mean count of all classes (14,158 instances).

2. Oversampling the Minorities



All other classes are synthetically boosted to 1.2 times the mean count (16,989 instances).



Tuning the Engine: Core Model Hyperparameters

Setting the fundamental parameters that control the model's learning process and structure.

Objective & Metric

- **`objective`**: `multiclass` (for 10 distinct classes)
- **`num_class`**: `10`
- **`metric`**: `multi_logloss` (primary loss function for optimization during training)

Learning Control

- **`learning_rate`**: `0.05` (A balanced step size for each boosting round)

Tree Structure Controls

- **`num_leaves`**: `80` (Balances model complexity and predictive power)
- **`max_depth`**: `-1` (No limit on depth; complexity is controlled by `num_leaves`)
- **`min_data_in_leaf`**: `60` (A regularization measure to prevent creating leaves for very small, noisy sample groups)

GPU Optimization

- **`max_bin`**: `255` (Optimal value for processing speed on GPU hardware)

Validation Setup

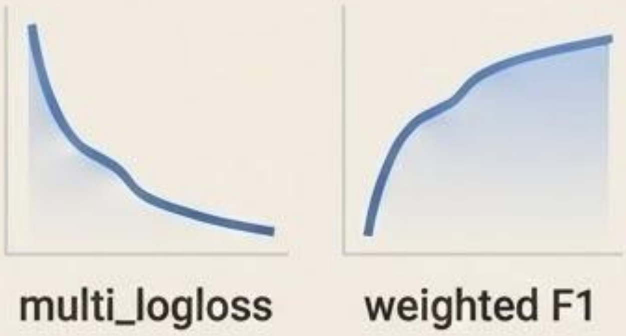
INTERNAL SPLIT

- Final training data split 80/20 for validation.

EARLY STOPPING

- Training halts if no improvement for 100 rounds.

METRICS



- Monitored multi_logloss and weighted F1 score throughout training.

EXPERIMENTAL RESULTS

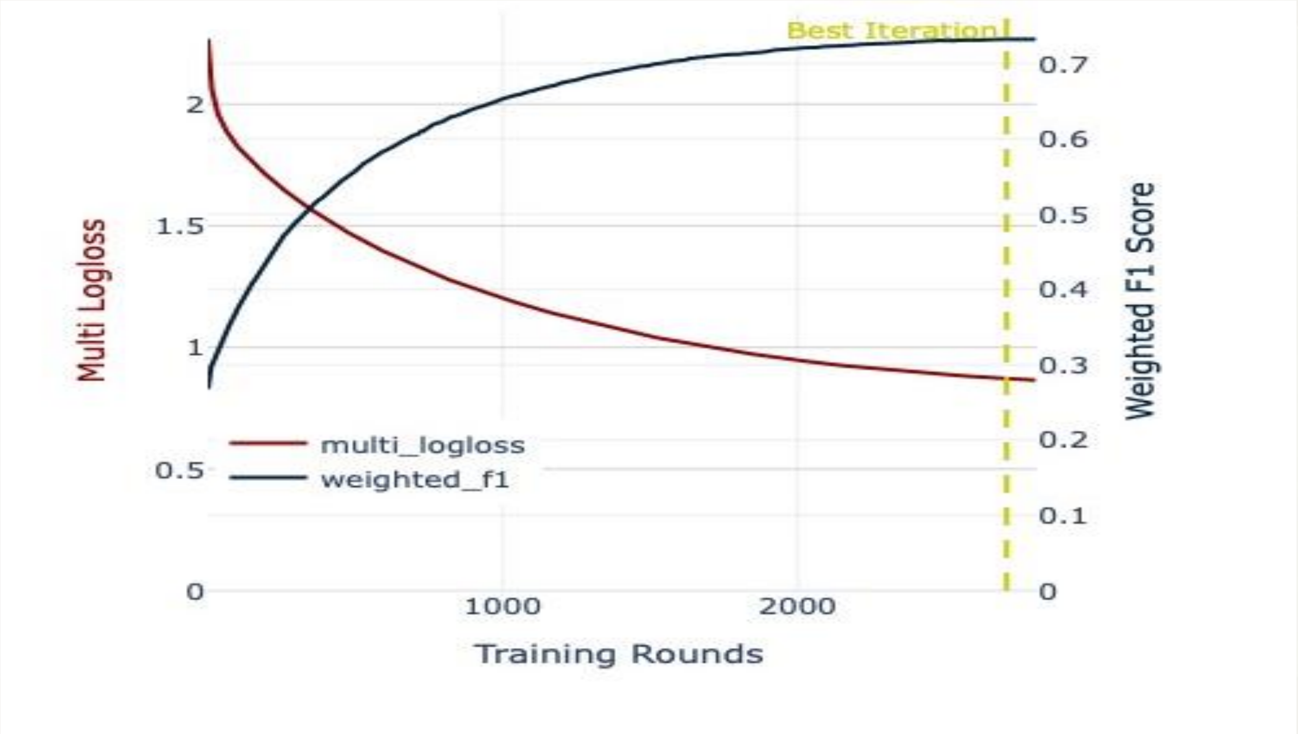
Reaching Peak Performance: A Balance of Learning and Generalization

The model was trained for up to 5,000 rounds with an early stopping condition to halt training if validation scores did not improve for 100 consecutive rounds. This ensures the model captures underlying patterns without overfitting to the training data.

Weighted F1 Score

0.733

This custom evaluation metric accounts for class imbalance, providing a robust measure of performance.



Multi-Logloss

~0.87

Shows the model's predictive confidence.

Optimal Training Point

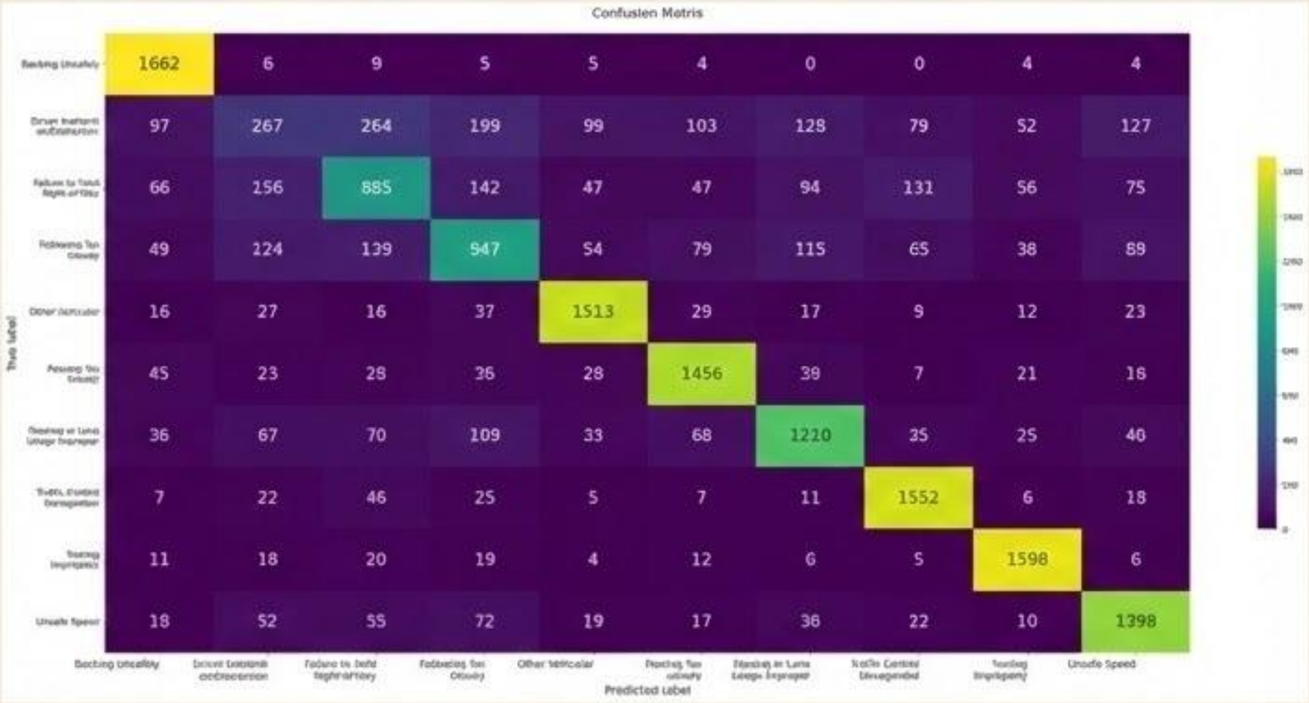
Best Iteration: 2707

Early stopping identified this as the ideal point, preventing performance degradation from further training.

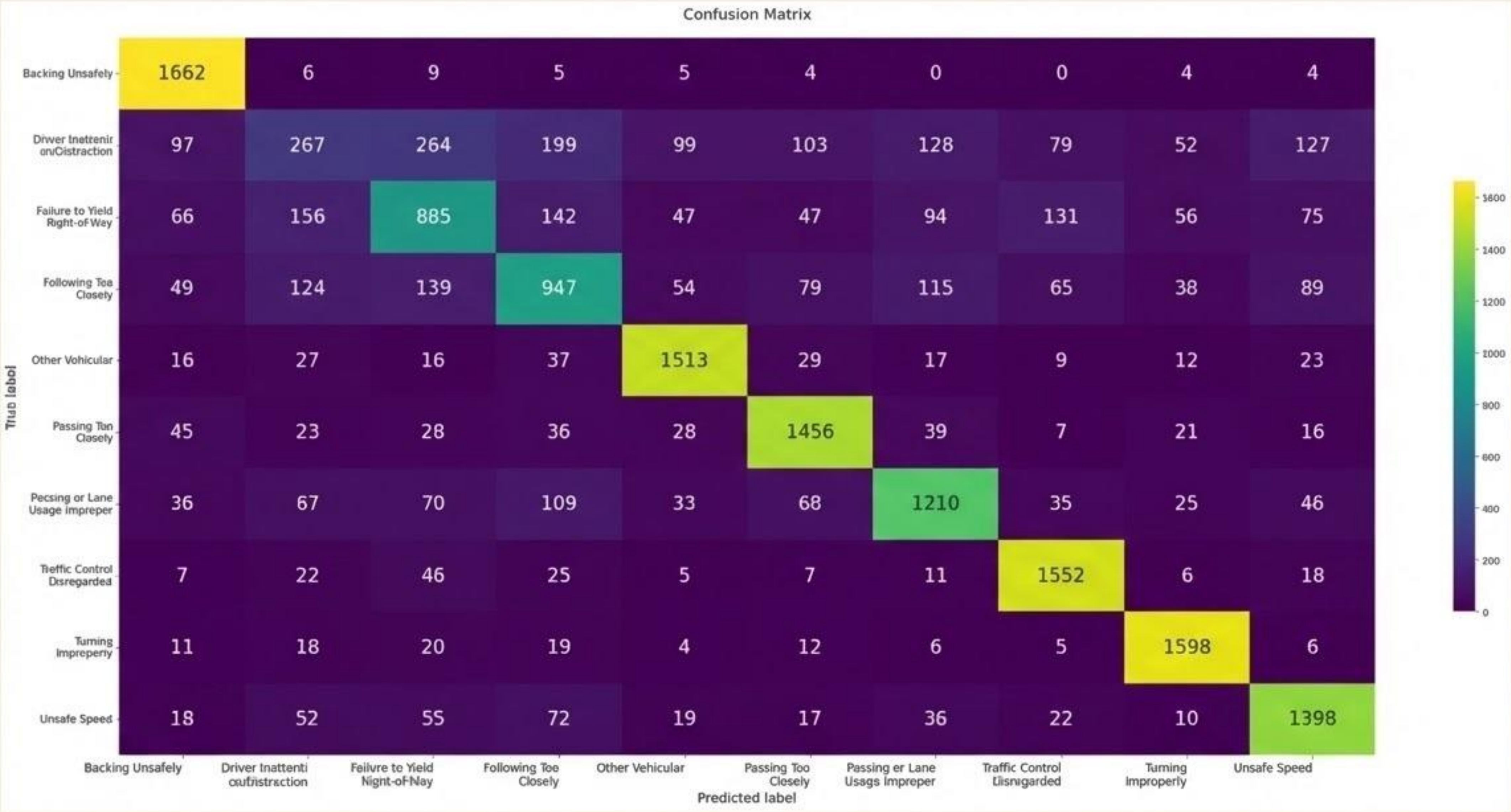
Confusion Matrix Overview

Key Takeaways

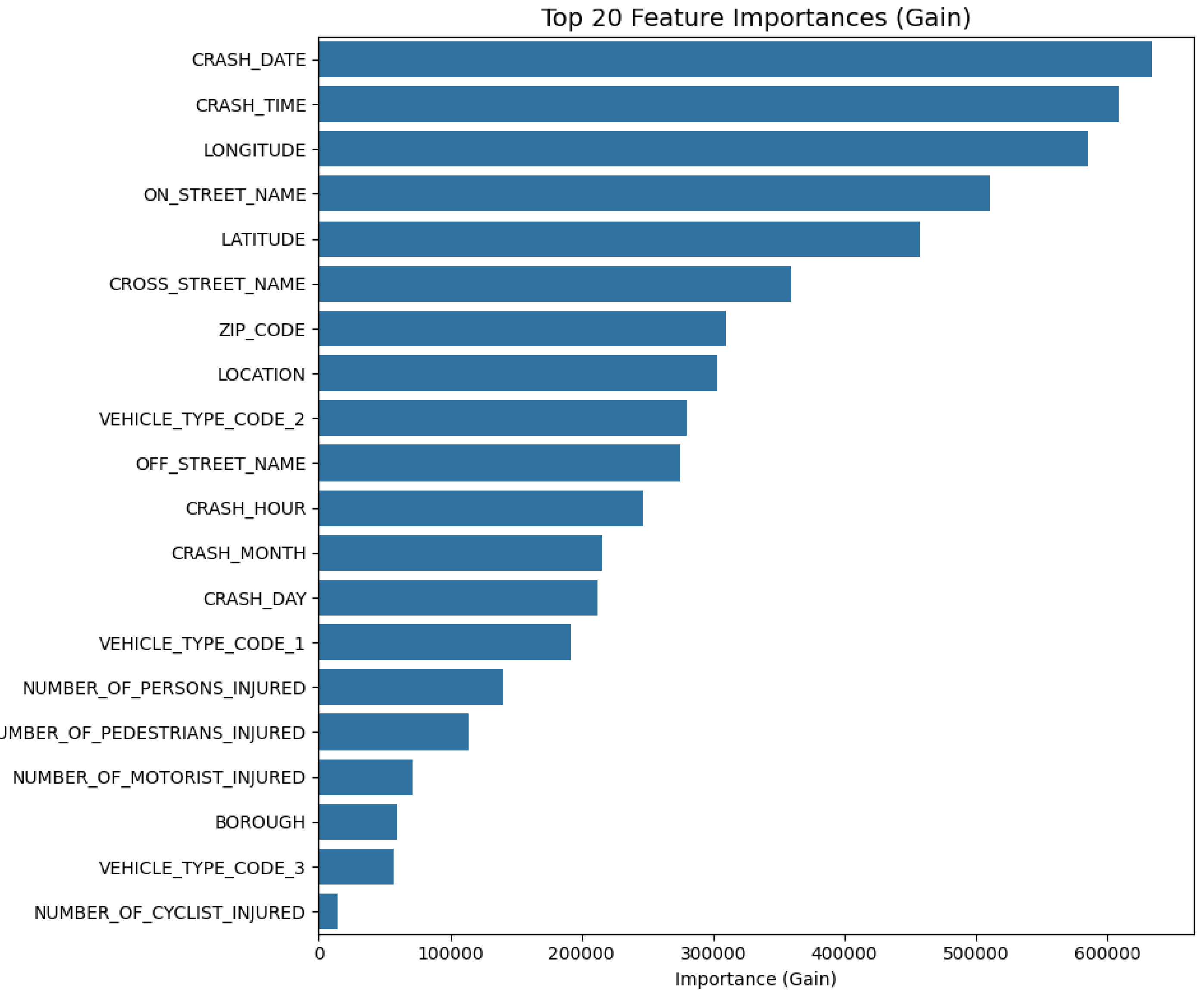
- **Clear Separation:** Distinct "Physical Maneuver" classes are well-clustered.
- **Behavioral Confusion:** High confusion exists among vague behavioral categories.
- **The Noise Source:** Driver Inattention is frequently mislabeled as other types.



Confusion Matrix Overview



Temporal and Geospatial Data are the Primary Predictors



1. Temporal Features (Top Predictors)

`CRASH_DATE`
`CRASH_TIME`

The model's strongest signals. Accident causes are heavily dependent on predictable temporal patterns, such as rush hour congestion versus late-night speeding.

2. Geospatial Features

`LONGITUDE`
`ON STREET NAME`
`LATITUDE`

High importance scores confirm that specific road geometries (e.g., complex intersections, as identified by SHAP analysis) are powerful predictors for maneuver-based incidents like "Turning Improperly".

3. Vehicular & Contextual Features

`VEHICLE TYPE CODE 1`
`CROSS STREET NAME`

The type of vehicle involved plays a key role, particularly for differentiating "Other Vehicular" causes, suggesting certain vehicles are prone to specific incident types.

Objective Actions Are Clearly Identified; Subjective States Are Not

The Resampling Success

High accuracy on minority classes like 'Backing Unsafely' (97.8%) validates our resampling strategy.

By oversampling minority classes to **1.2x the average count**, the model was able to learn their distinct patterns effectively.

Per-Class Prediction Accuracy Table

Contributing Factor Class	Accuracy	Performance Tier
Backing Unsafely	**97.8%**	[High]
Turning Improperly	**94.1%**	[High]
Traffic Control Disregarded	**91.3%**	[High]
Other Vehicular	89.1%	[High]
Passing Too Closely	85.7%	[High]
Unsafe Speed	82.3%	[High]
Passing or Lane Usage Improper	71.2%	[Moderate]
Following Too Closely	55.7%	[Moderate]
Failure to Yield Right-of-Way	52.1%	[Moderate]
Driver Inattention/Distraction	**18.9%**	[Low]

The Inattention Problem

The model's near-random performance on the majority class ('Driver Inattention/Distraction') indicates a severe lack of distinguishing features in the data. This suggests the label is either too ambiguous ambiguous or its data signature is indistinguishable from other behavioral causes.

Global Insights (XAI): What Features Matter Most?

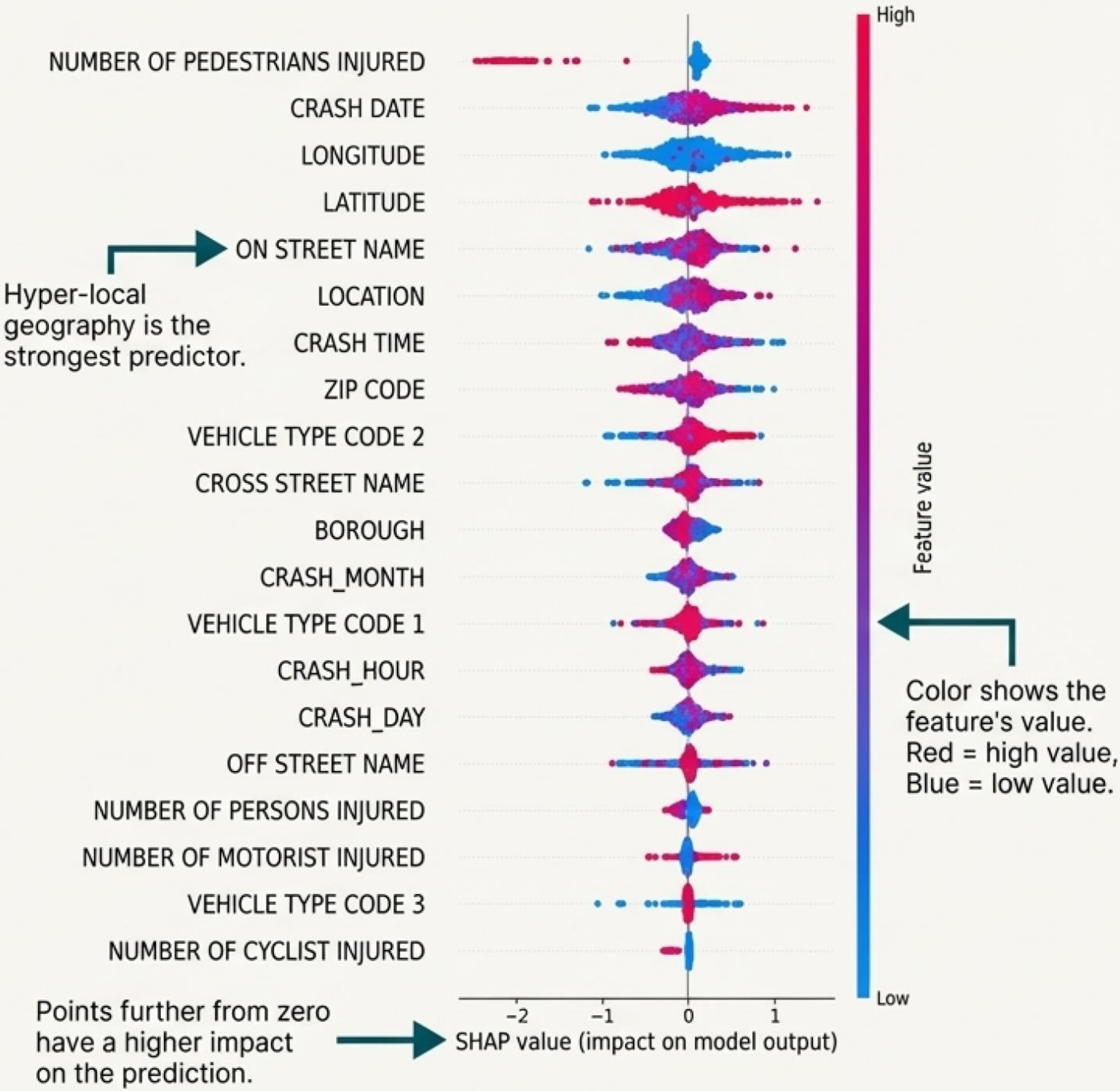
SHAP analysis reveals that location and time are the most powerful predictors of accident etiology.

Key Drivers:

- **Hyper-Local Geography:** `ON STREET NAME', `LONGITUDE', `LATITUDE' are consistently top factors, proving accidents are not random.

- **Precise Timing:** `CRASH DATE' and `CRASH TIME' are highly influential, capturing temporal trends and daily patterns.

- **Context:** `VEHICLE TYPE CODE` also provides a strong signal for predicting the cause.

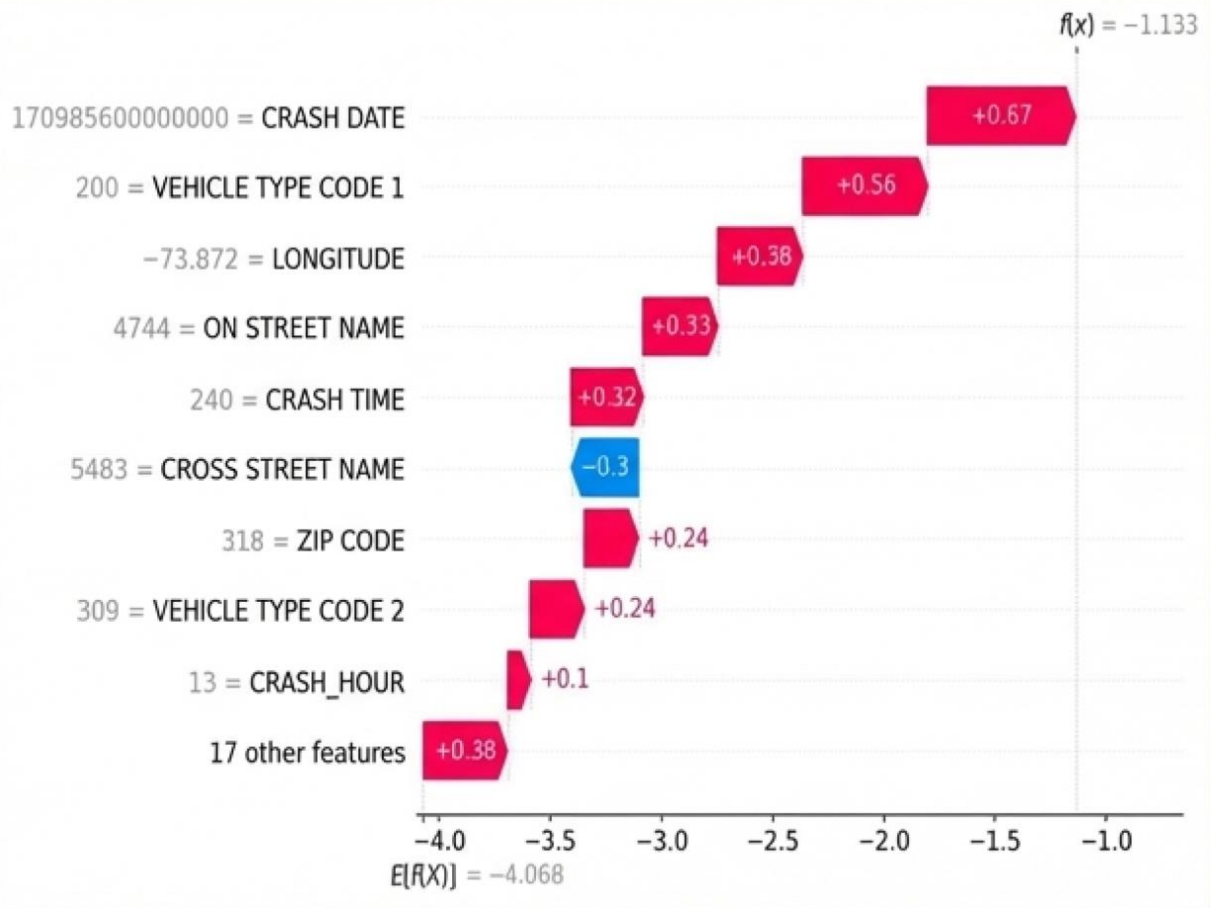


Local Insights (XAI): Anatomy of a Single Crash

XAI tools can dissect any individual prediction to explain its reasoning.

This SHAP “waterfall” plot explains why one specific incident was classified as “Traffic Control Disregarded.”

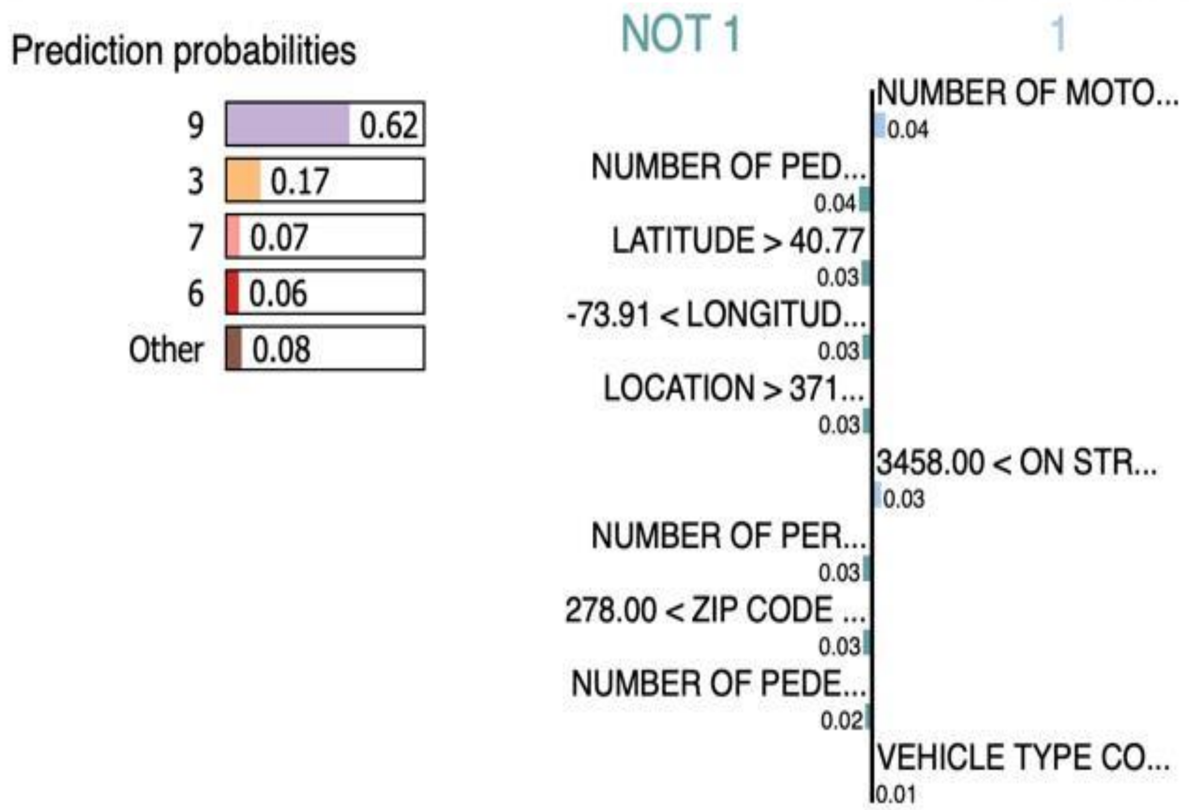
- **Driving Factors (Red):** The specific `CRASH DATE`, `VEHICLE TYPE CODE 1`, and `ON STREET NAME` strongly pushed the model toward this prediction.
- **Countervailing Factors (Blue):** The `CROSS STREET NAME` slightly reduced the probability.



POLICIES & SUMMARY

Application #1: Transforming 'Unknowns' into Actionable Data

A significant portion of the original dataset (**over 70,000 cases**) was labeled with the unhelpful contributing factor **"Unspecified."** We used our trained model to re-classify these cases.

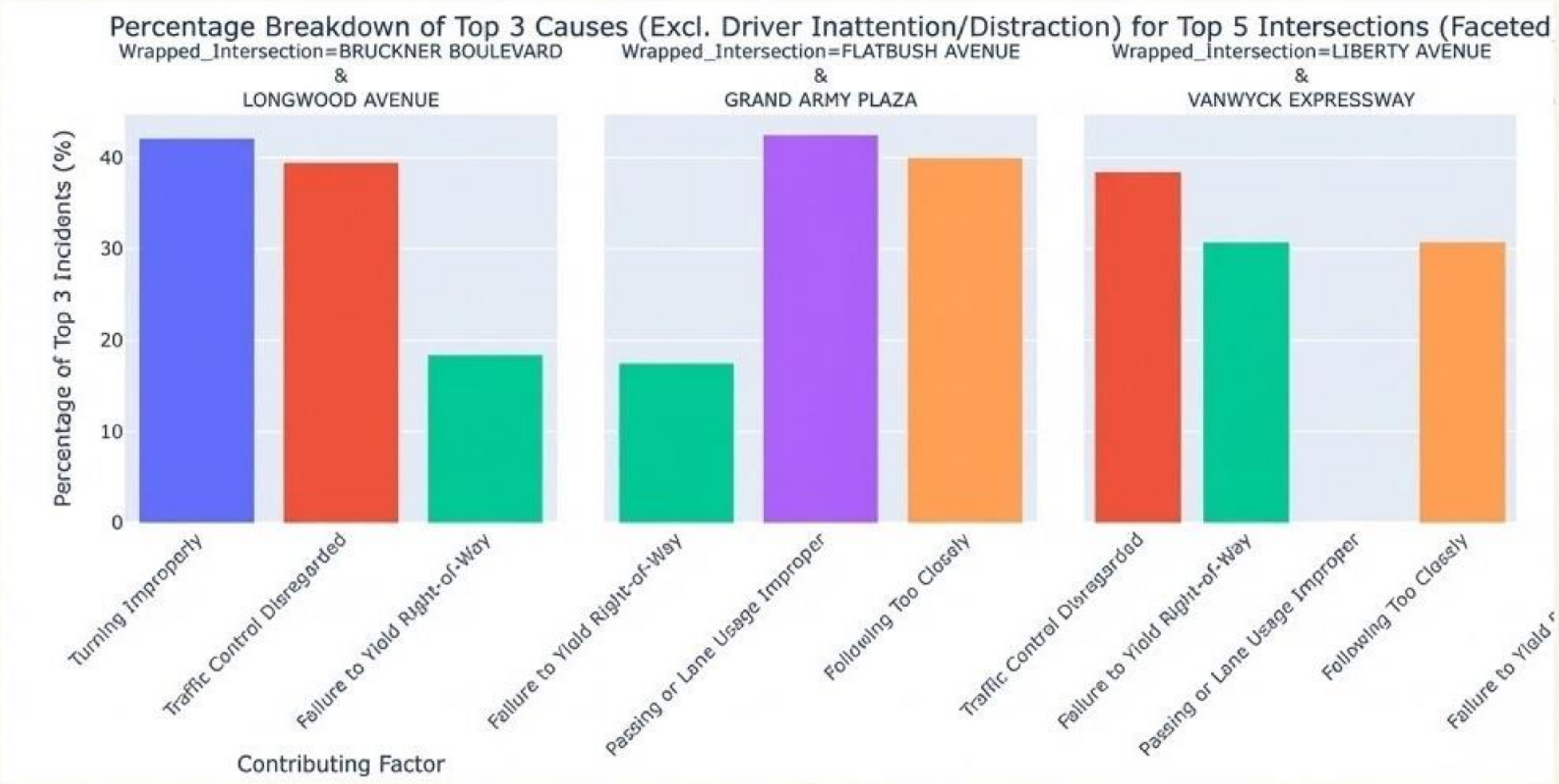


Feature	Value
NUMBER OF MOTORIST KILLED	0.00
NUMBER OF PEDESTRIANS KILLED	0.00
LATITUDE	40.88
LONGITUDE	-73.90
LOCATION	43189.00
ON STREET NAME	3471.00
NUMBER OF PERSONS KILLED	0.00
ZIP CODE	385.00
NUMBER OF PEDESTRIANS INJURED	0.00
VEHICLE TYPE CODE 3	74.00

This dramatically **improved** the dataset's utility, turning previously useless records into a valuable resource for further analysis.

Application #2: Generating Policy-Oriented Predictions for NYC Streets

By applying the model across New York City, we can predict the most likely contributing factors on different streets, providing targeted insights for safety interventions.



Actionable Insights:

This analysis can guide street-specific decisions on:

- Targeted traffic safety enforcement;
- Infrastructure improvement priorities;
- Public awareness campaigns

High-Priority Intersections

Causes & Suggested Actions for the Top 5 Locations

Bruckner Blvd & Longwood Ave
THE BRONX

- **Causes:** Improper turning, ignored signs, failure to yield.
- **Action:** Stricter monitoring and clearer turn signals to cut illegal turns.

Grand Army Plaza & Flatbush Ave
BROOKLYN

- **Causes:** Yielding issues, bad lane usage, tailgating.
- **Action:** Improve lane markings and signs for safer merging in the circle.

Van Wyck Expy & Liberty Ave
QUEENS

- **Causes:** Ignored signals, failure to yield, tailgating.
- **Action:** Use automated cameras and high-viz signals to reduce light-running.

Linden Blvd & Van Wyck Expy
QUEENS

- **Causes:** Failure to yield, improper passing, tailgating.
- **Action:** Add lane separators and countdown signals to prevent sudden changes.

Major Deegan Expy & W Fordham Rd
THE BRONX

- **Causes:** Improper turning, tailgating, failure to yield.
- **Action:** Adjust ramp layout and increase enforcement on aggressive turning.

Conclusion: What We Learned

The Signal: Metadata is excellent for identifying **WHERE** and **WHEN** crashes involving specific physical maneuvers occur. The model successfully finds these environmental risk zones.

The Noise: Current metadata is insufficient to distinguish the **WHY** of cognitive driver behaviors. These subjective states get lost in the noise of similar contextual data.

The True Value: The model is a powerful tool for **urban planning and policy-making**, not for assigning driver-specific liability.

The Next Frontier: Bridging the Gap Between Correlation and Causation

To move beyond predicting risk zones and begin to understand the cognitive 'why,' the next generation of models will require richer data sources.

Text from Officer Narratives

Apply Natural Language Processing (NLP) to extract nuanced details from written reports.

Vehicle Telematics

Incorporate data on speed, acceleration, and braking patterns preceding a crash.

Driver-Facing Camera Data

Provide direct visual evidence of distraction, fatigue, or other cognitive states.

GITHUB REPOSITORY

github.com/utki007/traffic-accident-ml-classifier

Thank You

Questions & Discussion